

Credit Card Approval Prediction

1. INTRODUCTION

1.1 Project Overview

The **Credit Card Approval Prediction** project is a machine learning-based system designed to predict whether a credit card application should be approved or rejected based on an applicant's personal and financial information. The system analyzes historical data, preprocesses the input, and uses a trained machine learning model to generate accurate predictions. By automating the credit evaluation process, the project helps reduce manual effort, improve decision-making accuracy, minimize processing time, and support financial institutions in making faster and more reliable credit card approval decisions through an easy-to-use web application.

1.2 Purpose

- Reduce manual effort in evaluating applications.
 - Improve approval accuracy.
 - Minimize financial risk.
 - Speed up the approval process.
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2. IDEATION PHASE

2.1 Problem Statement

Banks receive thousands of credit card applications daily. Manual verification is time-consuming and may lead to inconsistent decisions. An intelligent prediction system is needed to determine whether an applicant should be approved based on financial and

personal information.

2.2 Empathy Map Canvas

Include four sections:

- **Says:** "I need quick approval."
- **Thinks:** "Will my application be accepted?"
- **Does:** Submits financial and personal details.
- **Feels:** Hopeful, anxious, or frustrated.

2.3 Brainstorming

Discuss ideas such as:

- Using Machine Learning.
 - Predicting approval automatically.
 - Data preprocessing.
 - Feature engineering.
 - Building a web interface.
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3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Applicant registers → Enters details → Data validation → ML model predicts → Result displayed.

3.2 Solution Requirement

Functional Requirements

- User inputs applicant details.
- Validate input.

- Predict approval.
- Display result.

Non-functional Requirements

- Fast response.
- High accuracy.
- Secure.
- User-friendly interface.

3.3 Data Flow Diagram

Applicant



Input Form



Preprocessing



ML Model



Prediction



Approval / Rejection

3.4 Technology Stack

- Python
- Flask
- HTML
- CSS
- JavaScript
- Scikit-learn
- Pandas

- NumPy
 - Joblib
 - VS Code
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4. PROJECT DESIGN

4.1 Problem Solution Fit

Explain how the ML model solves the delay and inconsistency of manual credit card approval.

4.2 Proposed Solution

Describe the workflow:

- User enters applicant details.
- Data is preprocessed.
- ML model predicts approval.
- Decision displayed.

4.3 Solution Architecture

User



Web Application



Preprocessing



Machine Learning Model



Prediction

↓
Result

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activity
Week 1	Requirement Analysis
Week 2	Dataset Collection
Week 3	Data Preprocessing
Week 4	Model Training
Week 5	Model Evaluation
Week 6	Flask Integration
Week 7	Testing

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Include:

- Response Time
 - Prediction Accuracy
 - Input Validation
 - Concurrent Requests
 - Memory Usage
 - CPU Usage
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7. RESULTS

7.1 Output Screenshots

Insert screenshots of:

- Home Page
- Login Page (if available)
- Input Form
- Prediction Result
- Approved Output
- Rejected Output

8. ADVANTAGES & DISADVANTAGES

Advantages

- Faster approval process.
- Reduces manual work.
- High prediction accuracy.
- Consistent decisions.
- Easy to use.
- Scalable.

Disadvantages

- Depends on dataset quality.
 - Biased data may affect predictions.
 - Requires periodic retraining.
 - Cannot replace human judgment in exceptional cases.
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9. CONCLUSION

Summarize that the Credit Card Approval Prediction system successfully predicts approval decisions using machine learning, improving efficiency, reducing manual effort, and supporting faster, more consistent decision-making.

10. FUTURE SCOPE

- Improve prediction accuracy using deep learning.
 - Integrate real-time banking APIs.
 - Add fraud detection.
 - Deploy on cloud platforms.
 - Develop Android/iOS applications.
 - Enable continuous model retraining.
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11. APPENDIX

Include:

- Dataset description
- Input features
- Test cases
- Functional testing report
- Performance testing report
- References
- Source code snippets (optional)
- User manual